

STANDARD CELL LIBRARY DESIGN AND CHARACTERIZATION (SKY130)

Digital Design II - Spring 2026

Project 1 Report

Mohamed El-Refai (mohamedref11@aucegypt.edu) - 900222334

Haya Shalaby (hayashalaby@aucegypt.edu) - 900222031

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1. INTRODUCTION

The report documents the design and SPICE level characterization of a 13 cell standard cell library on the SKY130 130nm open source process node. We submitted a set of Non-Linear Delay Model (NLDM) timing tables, which is the format used by EDA tools to perform static timing analysis without resimulating every path at the transistor level.

The project was split into two parallel tracks. The circuit design track covered transistor sizing, structural SPICE netlist writing, and library compilation into a single file. The automation track covered testbench templating, ngspice batch execution, output parsing, and NLDM matrix generation. Both tracks were then integrated: the completed cell library was fed into the automation script, which ran the full 7x7 characterization sweep across all 13 cells.

2. TARGET LIBRARY

The library contains 13 cells across four logic functions:

Cell Type	Function	Drive Strengths	Total Cells
Inverter	$Y = \bar{A}$	invx1, invx2, invx4, invx8	4
NAND2	$Y = \overline{A \cdot B}$	nand2x1, nand2x2, nand2x4	3
NOR2	$Y = \overline{A + B}$	nor2x1, nor2x2, nor2x4	3
MAJ3	$Y = \overline{AB + BC + AC}$	maj3x1, maj3x2, maj3x4	3

All cells were characterized under the Typical-Typical (TT) process corner at 25°C and 1.8V supply, as specified.

3. CELL SIZING

3.1 Base Sizing (x1)

Transistor widths for the x1 base cells were derived from the SKY130 Kp measurement performed in Tutorial 2. Kp is the ratio of effective channel resistance between a PMOS and NMOS transistor of identical dimensions:

$$Kp = Tp / Tn = 2.943e-10 / 1.240e-10 = 2.372$$

We measured this by simulating the charge and discharge of a 25fF capacitor through a minimum size PMOS and NMOS respectively then extracting the 50% crossing times.

The minimum NMOS width in SKY130 is set by the design rules at $W_{min} = 0.42 \mu m$ with a fixed channel length of $L_{min} = 0.15 \mu m$ throughout and to achieve symmetric rise and fall times in the inverter we had to make the pull up and pull down resistances match, so the PMOS width is set to:

$$W_p = W_n \times K_p = 0.42 \times 2.372 \approx 1.00 \mu\text{m (rounded)}$$

This gives the x1 inverter sizing: NMOS $W = 0.42 \mu\text{m}$, PMOS $W = 1.00 \mu\text{m}$, $L = 0.15 \mu\text{m}$ for both.

3.2 Series Stack Compensation

For gates with transistors in series, the effective resistance of the stack is higher than a single device and to compensate the transistor widths in series configurations are scaled up:

NAND2: The pull down network has two NMOS in series. Worst case PDN has both in series, so NMOS width is doubled: $W_n = 0.42 \times 2 = 0.84 \mu\text{m}$. The PMOS are in parallel so they remain at base width: $W_p = 1.00 \mu\text{m}$.

NOR2: The PUN network has two PMOS in series. PMOS width is doubled: $W_p = 1.00 \times 2 = 2.00 \mu\text{m}$. The NMOS are in parallel so they remain at base width: $W_n = 0.42 \mu\text{m}$.

MAJ3: The MAJ3 PUN network implements $(A+B) \cdot (B+C) \cdot (A+C)$, three series groups each containing a parallel pair. In the worst case, only one transistor conducts per group, giving three devices in series. PMOS width is thus tripled: $W_p = 1.00 \times 3 = 3.00 \mu\text{m}$. The PUN network implements $AB + BC + AC$, three parallel branches of two NMOS in series. The worst case is a single active branch with two in series, so NMOS width is doubled: $W_n = 0.42 \times 2 = 0.84 \mu\text{m}$.

This worst case stack depth sizing produces more balanced timing across input combinations.

3.3 Drive Strength Scaling

Higher drive strength variants are made by multiplying all transistor widths by the drive strength factor N . For a cell sized at x1 widths W_n and W_p , the xN variant gives widths $N \times W_n$ and $N \times W_p$ respectively, with $L = 0.15 \mu\text{m}$ unchanged throughout. The series compensation factors are already integrated into the x1 sizing and so are preserved by the uniform scaling.

Final sizing summary:

Cell	NMOS W	PMOS W	L
invx1	0.42 μm	1.00 μm	0.15 μm
invx2	0.84 μm	2.00 μm	0.15 μm
invx4	1.68 μm	4.00 μm	0.15 μm
invx8	3.36 μm	8.00 μm	0.15 μm
nand2x1	0.84 μm	1.00 μm	0.15 μm
nand2x2	1.68 μm	2.00 μm	0.15 μm

nand2x4	3.36 μm	4.00 μm	0.15 μm
nor2x1	0.42 μm	2.00 μm	0.15 μm
nor2x2	0.84 μm	4.00 μm	0.15 μm
nor2x4	1.68 μm	8.00 μm	0.15 μm
maj3x1	0.84 μm	3.00 μm	0.15 μm
maj3x2	1.68 μm	6.00 μm	0.15 μm
maj3x4	3.36 μm	12.00 μm	0.15 μm

4. CIRCUIT IMPLEMENTATION

All cells were implemented as transistor-level SPICE subcircuits using SKY130 native device models: sky130_fd_pr__nfet_01v8 for NMOS and sky130_fd_pr__pfet_01v8 for PMOS. These are the 1.8V core devices from the SKY130 foundry primitive library.

The pin order for each cell type follows the convention expected by the characterization automation:

Inverter: (A Y VDD VSS)
NAND2/NOR2: (A B Y VDD VSS)
MAJ3: (A B C Y VDD VSS)

The MAJ3's pull-up network implements $(A+B) \cdot (B+C) \cdot (A+C)$ using three series connected PMOS pairs in parallel. The pull-down network implements $AB + BC + AC$ using three parallel branches of series NMOS pairs. A simple inverter that restores the non-inverted majority output, resulting in a non-inverting output: $Y = AB + BC + AC$.

All 13 subcircuit definitions are compiled into a single file: spice/stdcells.lib.spice.

5. CHARACTERIZATION METHODOLOGY

5.1 Characterization

Characterization is the process of pre-running SPICE simulations on each cell across various operating conditions, and packaging the results into lookup tables. EDA tools use these tables during static timing analysis to estimate path delays without running transistor-level simulation on the full design. The NLDM format stores delay as a 2D function of input transition time and output load capacitance, capturing the nonlinear dependence that a simple RC model cannot express.

5.2 NLDM Tables

Four timing quantities are extracted per cell:

cell_rise: Propagation delay for a rising output transition. Measured from when the input crosses 50% of VDD to when the output crosses 50% of VDD in the rising direction.

cell_fall: Propagation delay for a falling output transition. Same threshold definition in the falling direction.

rise_transition: Output rise slew time, measured from 20% VDD to 80% VDD on the output waveform.

fall_transition: Output fall slew time, measured from 80% VDD to 20% VDD on the output waveform.

The 50%/50% threshold for propagation delay isolates the cell's contribution from the input and output ramp times. The 20%/80% thresholds for slew capture the linear portion of the transition and exclude the slow tails near the supply.

5.3 Sweep Vectors

Each table is 7×7 , covering the following input transition and load capacitance vectors:

Input transition times (ns): 0.0100, 0.0231, 0.0531, 0.1225, 0.2823, 0.6507, 1.5000

Output load capacitances (pF): 0.0005, 0.0013, 0.0035, 0.0094, 0.0249, 0.0662, 0.1758

This gives 49 simulations per cell and 637 simulations total across the full library.

5.4 Automation

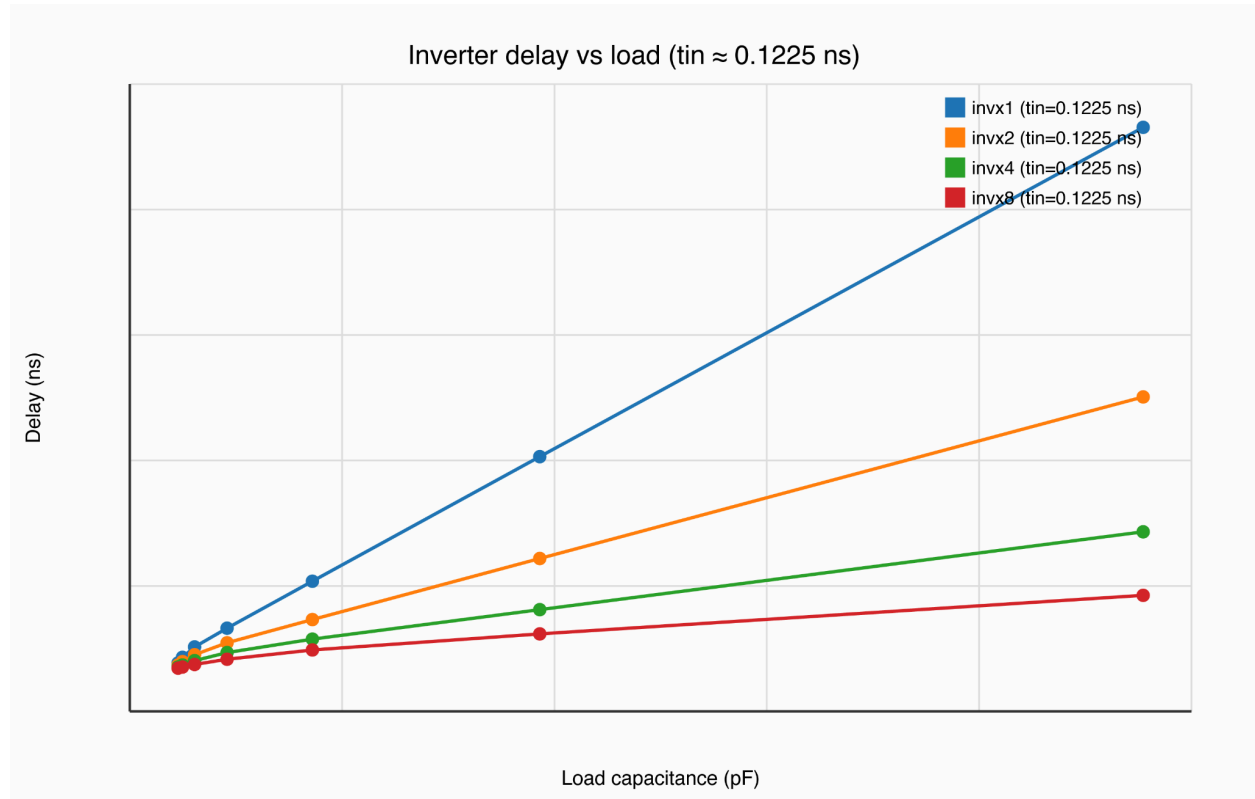
A Python script (scripts/run_characterization.py) automates the full sweep. For each (cell, tin, Cload) combination it generates a parametrized ngspice testbench from a Jinja2 template, runs ngspice in batch mode, and parses the .measure results from the simulation log. The results are written to JSON files in results/nldm/, one per cell, each containing the four 7×7 matrices and run metadata.

For multi-input cells (NAND2, NOR2, MAJ3), the non-switching inputs are held at non-controlling logic levels, NAND2 holds B at VDD, NOR2 holds B at VSS, and MAJ3 holds B at VSS and C at VDD. This ensures that the switching input A controls the output and produces valid rising and falling arcs for measurement.

6. NLDM RESULTS

NLDM tables placed in appendix as well as an external excel file which is also in the repo.

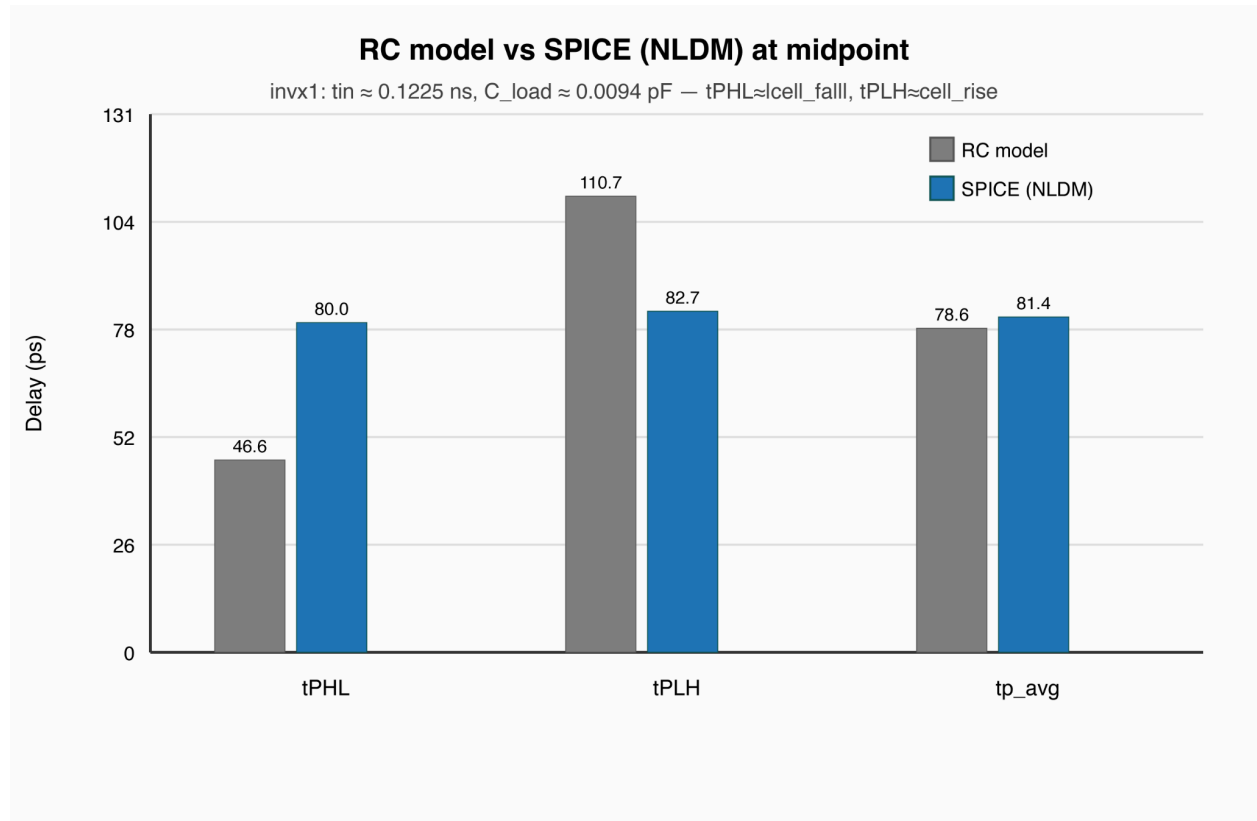
7. INVERTER FAMILY: DELAY VS. LOAD



Propagation delay vs output load for invx1–invx8 at $t_{in} \approx 0.1225$ ns (average of rise/fall delays from NLDM)

The trend is that delay increases linearly with load for small loads, and that each higher drive strength variant shifts the curve downward so invx2 should show roughly half the delay of invx1 at any given load, invx4 a quarter, and so on. This is the direct consequence of halved resistance at doubled width.

8. RC MODEL VS. SPICE COMPARISON



midpoint t_{in} , C_{load} , invx1, RC vs NLDM

8.1 The RC Model

The first-order RC delay model approximates a CMOS gate by treating each transistor as a resistor charging or discharging the load capacitance. The propagation delay is given by:

$$tp = 0.69 \times R_{eff} \times C_{load}$$

where 0.69 is the natural logarithm of 2, arising from the 50% crossing threshold. R_{eff} is the effective channel resistance of the conducting transistor.

The effective resistances for the minimum-size SKY130 transistors were extracted in Tutorial 2 by measuring the time constant of a known capacitor ($C = 25$ fF) being charged or discharged through each device:

$$R_{NMOS} = T_N / (0.69 \times C) = 1.240e-10 / (0.69 \times 25e-15) = 7.19 \text{ k}\Omega$$
$$R_{PMOS} = T_P / (0.69 \times C) = 2.943e-10 / (0.69 \times 25e-15) = 17.06 \text{ k}\Omega$$

These correspond to transistors with $W = 0.42 \text{ }\mu\text{m}$ and $W = 1.00 \text{ }\mu\text{m}$ respectively, which are the exact dimensions used in the invx1 cell.

8.2 RC Delay Calculation at the Midpoint

The comparison is performed at the center of the NLDM table: input transition $t_{in} = 0.1225$ ns and load capacitance $C_{load} = 0.0094$ pF.

$$\begin{aligned} t_{PHL} (RC) &= 0.69 \times R_{NMOS} \times C_{load} \\ &= 0.69 \times 7190 \times 0.0094 \times 10^{-12} \\ &= 46.6 \text{ ps} \end{aligned}$$

$$\begin{aligned} t_{PLH} (RC) &= 0.69 \times R_{PMOS} \times C_{load} \\ &= 0.69 \times 17060 \times 0.0094 \times 10^{-12} \\ &= 110.6 \text{ ps} \end{aligned}$$

$$tp_{avg} (RC) = (t_{PHL} + t_{PLH}) / 2 = 78.6 \text{ ps}$$

8.3 SPICE Result at the Midpoint

At the NLDM grid point $t_{in} = 0.1225$ ns, $C_{load} = 0.0094$ pF, $invx1$ gives $cell_rise = 0.082733$ ns and $cell_fall = 0.08001693$ ns. Converting to picoseconds and using the definitions above:

$$\begin{aligned} t_{PHL} (SPICE) &= 80.0 \text{ ps} \\ t_{PLH} (SPICE) &= 82.7 \text{ ps} \\ tp_{avg} (SPICE) &= 81.4 \text{ ps} \end{aligned}$$

8.4 Comparison and Discussion

Quantity	RC Model	SPICE NLDM	Error
tPHL (ps)	46.6	80.0	+71.7%
tPLH (ps)	110.6	82.7	-25.2%
tp_avg (ps)	78.6	81.4	+3.5%

The RC model splits pull-down vs pull-up with fixed R_n and R_p , while SPICE at this point gives similar rise and fall delays (~ 81 ps), so tp_{avg} still agrees within a few percent even though t_{PHL} and t_{PLH} do not track the RC split—consistent with finite input slew and non-linear device behavior.

The RC model captures the first-order dependence of delay on load capacitance reasonably well near the midpoint of the operating range, but has two fundamental limitations that the NLDM table addresses.

First, the RC model ignores input transition time entirely. The formula $t_p = 0.69 \times R \times C$ produces a single delay value regardless of how fast or slow the input switches. In reality, a slow input edge means the transistor spends a significant amount of time partially on, neither fully conducting nor fully off, which changes the effective charging behavior and therefore the output delay. The NLDM table captures this through its second dimension: across the seven input transition values from 0.01 ns to 1.5 ns, delay changes substantially, something the RC model cannot represent at all.

Second, the RC model linearizes what is a nonlinear relationship. It assumes the transistor behaves as a constant resistor throughout the switching event. In practice, the channel resistance varies with drain-to-source voltage as the transistor moves between saturation and linear regions. At small loads, where switching is fast and the transistor spends most of its time in saturation, the RC approximation is relatively accurate. At large loads, the longer switching time means the transistor spends more time in the linear region where its behavior deviates more from a simple resistor, and the model becomes less accurate.

The NLDM table sidesteps both issues by directly measuring delay from SPICE at each grid point, capturing all the nonlinear device physics implicitly. The RC model remains useful as a fast hand-calculation tool and for intuition about how sizing affects delay, but it is not accurate enough for sign-off timing analysis.

9. CONCLUSION

A 13-cell standard cell library was designed from scratch for the SKY130 process and characterized using an automated ngspice sweep. Transistor sizing was grounded in the measured K_p of the process (2.372), with series stack compensation applied based on worst-case stack depth rather than nominal depth, a more rigorous approach that improves timing balance across input combinations.

The NLDM characterization flow produces four 7×7 timing tables per cell covering cell_rise, cell_fall, rise_transition, and fall_transition across a range of input slew and output load conditions. The RC model comparison at the midpoint demonstrates both the utility and the limitations of first-order analytical models: the RC model gives a reasonable estimate of delay but cannot capture the dependence on input transition time, which is a significant portion of what the NLDM format encodes.

10. AI USE AND PRACTICAL ISSUES

We used AI tools during this project for debugging and environment troubleshooting but the technical content of cell sizing, spice netlists, characterization methodology, choice of sweep vectors, interpretation of NLDM vs RC, and final numerical results was developed and verified by us using ngspice, the SKY130 PDK, and our own simulation outputs. AI suggestions were

reviewed, tested, and corrected when they did not match measured results or actual tool behavior. Also, much of our time was spent on integration not into writing the library on paper as ngspice batch runs with the SKY130 PDK were sensitive to library paths, working directory, include order, and ngspice version, errors often showed up only as missing or unresolved device models or parser failures (including interactions with PDK init / control syntax in batch mode), which required many iterative fixes. In addition, automation (Jinja2 testbenches, log parsing, and JSON assembly) had to be debugged across failed runs, null NLDM entries, and corner cases in multi-input cells (biasing non-switching inputs and valid rise/fall arcs). Besides that, on the apple silicon we also hit environment issues (e.g. matching Python, optional plotting packages, and conda/pyenv layouts) that took alot of time. Together, these issues explain the long time compared to the results we have. Additionally, AI was used for productivity to put our NLDM produced tables into excel format for analysis and use in this report.

11. DIVISION OF WORK

Mohamed El-Refai

- Standard cell library (design side): transistor-level SPICE subcircuits for the 13 cells, pin conventions aligned with the characterization flow, and integration into spice/stdcells.lib.spice.
- Sizing support: applying Kp-based x1 sizing, series-stack compensation (NAND2 / NOR2 / MAJ3), and drive-strength scaling (x2 / x4 / x8) consistently across the library.
- Characterization automation: scripts/run_characterization.py workflow as Jinja2 testbench generation, ngspice batch execution, log / .measure parsing, and JSON NLDM output under results/nldm/, debugging of PDK paths, batch-deck issues, and failed runs until tables were complete with Haya helping with the run on her windows machine.
- Analysis tooling: plotting scripts (e.g. inverter delay vs load, RC vs SPICE comparison at the NLDM midpoint) and related environment/setup troubleshooting on macOS / conda.

Haya Shalaby

- Standard cell library (design side): schematic-level / structural decisions for NAND2, NOR2, MAJ3, and inverter variants, and cross-checking logic and stacking against the sizing rules in the report.
- Characterization methodology: in section 5 narrative as NLDM definitions (delay vs slew, 50% / 20–80% thresholds), 7×7 sweep vectors, simulation count, and multi-input biasing (NAND2 / NOR2 / MAJ3) for valid timing arcs.
- NLDM deliverables: taking the final NLDM numbers (from JSON / simulation logs) and assembling them into nldm_tables.xlsx in four tables per cell, row/column labels (input transition × load capacitance), formatting for the written appendix and instructor review.
- Repository / integration: coordinating Git branches (e.g. NLDM vs design branch), merging characterization results with the report structure, and scripts/generate_nldm_excel.py (or equivalent flow) so the Excel book stays consistent with results/nldm/*.json.

APPENDIX: NLDM TABLES

INVX1

Cell Rise Delay (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.009989	0.014588	0.026216	0.05623	0.134426	0.342493	0.89062
0.0231	0.013019	0.01742	0.029059	0.059219	0.137512	0.345623	0.893862
0.0531	0.018271	0.024313	0.035996	0.066176	0.144591	0.35278	0.901261
0.1225	0.02473	0.034375	0.051717	0.082733	0.161106	0.369371	0.918341
0.2823	0.032358	0.047137	0.074708	0.119898	0.199728	0.407776	0.957545
0.6507	0.041751	0.062782	0.10427	0.175876	0.287968	0.497227	1.047557
1.5	0.053307	0.082245	0.141059	0.248779	0.426495	0.705118	1.254938

Cell Fall Delay (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.010879	0.015303	0.026462	0.055217	0.130122	0.329442	0.858285
0.0231	0.01332	0.017739	0.029074	0.058033	0.133046	0.332412	0.861272
0.0531	0.01811	0.023842	0.035404	0.064559	0.139748	0.339202	0.868099
0.1225	0.02405	0.032938	0.049647	0.080017	0.155346	0.354931	0.883893
0.2823	0.030829	0.044297	0.069998	0.113933	0.191716	0.39129	0.920305
0.6507	0.037675	0.057536	0.09594	0.163184	0.272931	0.475806	1.004494
1.5	0.042069	0.070655	0.1268	0.226953	0.394748	0.669259	1.199808

Rise Transition Time (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.007125	0.01306	0.029379	0.073151	0.188146	0.49455	1.307672
0.0231	0.007676	0.013067	0.029383	0.073153	0.188146	0.49455	1.307672
0.0531	0.012661	0.016359	0.02991	0.073153	0.188146	0.49455	1.307672
0.1225	0.020455	0.026834	0.038478	0.074653	0.188146	0.49455	1.307672
0.2823	0.033757	0.044177	0.062467	0.093634	0.1908	0.49455	1.307671
0.6507	0.056986	0.072574	0.102532	0.150132	0.230985	0.497839	1.307672
1.5	0.103481	0.123946	0.167929	0.245953	0.363573	0.579164	1.310049

Fall Transition Time (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.005801	0.010819	0.024666	0.061845	0.159521	0.419787	1.110469

0.0231	0.005991	0.01082	0.024667	0.061844	0.159521	0.419786	1.110469
0.0531	0.009327	0.013129	0.025	0.061845	0.159521	0.419787	1.110469
0.1225	0.01554	0.021429	0.032388	0.063171	0.159521	0.419787	1.110469
0.2823	0.026489	0.035471	0.052129	0.080775	0.162313	0.419786	1.110469
0.6507	0.047631	0.060429	0.085181	0.128251	0.200873	0.423826	1.110469
1.5	0.091368	0.108679	0.143793	0.207504	0.314537	0.503668	1.114285

INVX2

Cell Rise Delay (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.008455	0.010887	0.016978	0.032175	0.071128	0.174409	0.448274
0.0231	0.011733	0.013928	0.019874	0.035158	0.074249	0.17761	0.451509
0.0531	0.016229	0.019966	0.027183	0.042284	0.081461	0.18494	0.458904
0.1225	0.022095	0.028068	0.039658	0.059434	0.098461	0.20198	0.476028
0.2823	0.02961	0.038618	0.056817	0.088382	0.138498	0.241615	0.515585
0.6507	0.040285	0.053022	0.079707	0.128703	0.208562	0.334411	0.607379
1.5	0.05661	0.074022	0.111515	0.183038	0.306972	0.507072	0.82142

Cell Fall Delay (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.009076	0.011343	0.017043	0.031195	0.067406	0.163416	0.418013
0.0231	0.011621	0.013816	0.019552	0.033906	0.070286	0.166378	0.421008
0.0531	0.015449	0.018911	0.02592	0.040342	0.076923	0.173158	0.427853
0.1225	0.019912	0.025326	0.03625	0.055593	0.092521	0.188903	0.443702
0.2823	0.024124	0.032365	0.049186	0.079261	0.1287	0.225474	0.480296
0.6507	0.026241	0.038449	0.063546	0.109372	0.186	0.310785	0.565161
1.5	0.021699	0.039163	0.075739	0.143572	0.260028	0.453124	0.763191

Rise Transition Time (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.005229	0.008146	0.016173	0.037747	0.09441	0.245403	0.646099

0.0231	0.006129	0.008427	0.016181	0.037742	0.094412	0.245402	0.646099
0.0531	0.010565	0.013134	0.018378	0.037757	0.094411	0.245403	0.646099
0.1225	0.017266	0.021191	0.028794	0.043845	0.094627	0.245403	0.646099
0.2823	0.02881	0.035144	0.047384	0.067824	0.107924	0.245567	0.646099
0.6507	0.049657	0.058764	0.07806	0.111545	0.163146	0.272056	0.646099
1.5	0.092337	0.103894	0.130776	0.183017	0.267717	0.39703	0.694927

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.004059	0.006456	0.013075	0.03086	0.077623	0.202227	0.532897
0.0231	0.004563	0.00659	0.013075	0.030863	0.077624	0.202227	0.532897
0.0531	0.007658	0.009989	0.014876	0.030917	0.077623	0.202227	0.532897
0.1225	0.013056	0.01669	0.023732	0.03684	0.07801	0.202226	0.532897
0.2823	0.022672	0.028269	0.039257	0.058259	0.09212	0.202809	0.532897
0.6507	0.04163	0.049595	0.065822	0.095061	0.143033	0.232571	0.533158
1.5	0.081602	0.092038	0.11486	0.157733	0.231281	0.351791	0.591367

INVX4

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.007507	0.008754	0.011951	0.019814	0.039337	0.090483	0.225773
0.0231	0.010707	0.011999	0.014949	0.022711	0.042346	0.093622	0.228981
0.0531	0.014654	0.016858	0.021517	0.029973	0.049478	0.100856	0.236324
0.1225	0.019876	0.023348	0.030765	0.044312	0.066713	0.117863	0.253382
0.2823	0.027179	0.032224	0.043404	0.064694	0.100213	0.157968	0.293017
0.6507	0.038943	0.045898	0.061669	0.093054	0.148483	0.238888	0.385805
1.5	0.059917	0.069057	0.090527	0.134603	0.216077	0.357262	0.581913

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.007834	0.008935	0.011776	0.018704	0.03586	0.080822	0.1998
0.0231	0.010174	0.011436	0.014224	0.021241	0.038586	0.083677	0.202713
0.0531	0.013096	0.015054	0.019365	0.027537	0.044968	0.090237	0.20938
0.1225	0.01597	0.019035	0.025763	0.038518	0.060269	0.105577	0.224842

0.2823	0.017438	0.022076	0.032317	0.051983	0.085749	0.141615	0.260678
0.6507	0.014447	0.02129	0.036451	0.065878	0.117569	0.204068	0.344386
1.5	0.000225	0.00974	0.031432	0.074427	0.151305	0.283523	0.499633

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.004123	0.005554	0.009496	0.020118	0.048038	0.122424	0.319833
0.0231	0.005508	0.006412	0.00963	0.020124	0.048036	0.122425	0.319833
0.0531	0.009493	0.010995	0.013871	0.021608	0.048036	0.122424	0.319833
0.1225	0.01577	0.018103	0.02295	0.031943	0.052161	0.122424	0.319833
0.2823	0.02658	0.030194	0.038112	0.052373	0.075593	0.130996	0.319833
0.6507	0.047218	0.052042	0.063501	0.086349	0.123936	0.182946	0.335323
1.5	0.09149	0.097009	0.111512	0.14375	0.203582	0.299111	0.446683

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.003055	0.004205	0.007259	0.01553	0.037265	0.095215	0.248998
0.0231	0.003941	0.004776	0.007376	0.01553	0.037268	0.095215	0.248998
0.0531	0.00669	0.007998	0.010852	0.017025	0.037269	0.095215	0.248998
0.1225	0.011434	0.013557	0.018006	0.026154	0.042089	0.095235	0.248997
0.2823	0.020303	0.023475	0.030281	0.042935	0.064173	0.10637	0.248997
0.6507	0.038793	0.043096	0.052702	0.071407	0.104055	0.158125	0.271218
1.5	0.079989	0.084866	0.097276	0.123429	0.171391	0.254274	0.389186

INVX8

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.007049	0.007652	0.009296	0.013418	0.023391	0.048825	0.115556
0.0231	0.010076	0.010857	0.012476	0.016358	0.026289	0.051856	0.118703
0.0531	0.013699	0.014927	0.017746	0.023342	0.033465	0.058974	0.125939
0.1225	0.018503	0.020401	0.024833	0.03372	0.049217	0.076066	0.142887
0.2823	0.025755	0.028424	0.034814	0.048223	0.072614	0.113179	0.182718
0.6507	0.038455	0.04205	0.050796	0.069633	0.105758	0.169324	0.271361
1.5	0.062918	0.067481	0.078965	0.104584	0.155193	0.249247	0.409321

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.007321	0.007887	0.009365	0.013036	0.021855	0.044336	0.103387
0.0231	0.009432	0.010132	0.01177	0.015399	0.024345	0.047006	0.106161
0.0531	0.011725	0.012802	0.015364	0.020681	0.030417	0.053213	0.112526
0.1225	0.013499	0.015163	0.019119	0.02732	0.042231	0.067986	0.127365
0.2823	0.013203	0.015682	0.021582	0.033903	0.056687	0.096026	0.162071
0.6507	0.00713	0.010749	0.019382	0.037455	0.071239	0.13104	0.230632
1.5	-0.01282	-0.00792	0.004103	0.029982	0.079128	0.167497	0.318805

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.003569	0.004262	0.006208	0.011397	0.025057	0.061466	0.158091
0.0231	0.005226	0.005595	0.006896	0.011388	0.025055	0.061462	0.158091
0.0531	0.008871	0.009688	0.011612	0.015013	0.025881	0.061464	0.15809
0.1225	0.014762	0.016064	0.019062	0.024886	0.035038	0.063759	0.158091
0.2823	0.024916	0.026861	0.031554	0.041144	0.057437	0.08425	0.162415
0.6507	0.045067	0.047503	0.053765	0.067862	0.094382	0.137152	0.206274
1.5	0.089278	0.09187	0.0991	0.117158	0.155345	0.224771	0.33201

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.002765	0.003285	0.004814	0.008891	0.019668	0.048418	0.124723
0.0231	0.003821	0.004169	0.005336	0.008922	0.019669	0.048418	0.124723
0.0531	0.006246	0.006972	0.008734	0.012264	0.020688	0.048417	0.124723
0.1225	0.010518	0.011729	0.01449	0.019949	0.029677	0.051648	0.124723
0.2823	0.018784	0.020511	0.024562	0.032842	0.047763	0.073151	0.131785
0.6507	0.036619	0.038857	0.044294	0.055855	0.077759	0.116476	0.180556
1.5	0.077588	0.079782	0.085983	0.101046	0.131446	0.187968	0.285473

NAND2X1

Cell Rise Delay (ns)

tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.013238	0.018139	0.030341	0.060831	0.13927	0.347441	0.898611
0.0231	0.015964	0.020594	0.032763	0.063572	0.142277	0.350585	0.90184
0.0531	0.022474	0.027546	0.039231	0.070055	0.149093	0.357642	0.90907
0.1225	0.031423	0.039494	0.055344	0.086116	0.165148	0.373986	0.925733
0.2823	0.043534	0.055876	0.080685	0.123778	0.203346	0.412002	0.964127
0.6507	0.061904	0.079301	0.116056	0.183193	0.292331	0.501243	1.052996
1.5	0.092978	0.116751	0.167946	0.266606	0.436604	0.710502	1.259885

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.01162	0.014958	0.023526	0.045665	0.103294	0.256612	0.663386
0.0231	0.013862	0.017237	0.025956	0.04826	0.105982	0.25934	0.66613
0.0531	0.018324	0.022766	0.03192	0.054372	0.112236	0.265668	0.67249
0.1225	0.023415	0.03023	0.043994	0.068929	0.126856	0.280378	0.687246
0.2823	0.028014	0.038278	0.059245	0.0969	0.161073	0.314447	0.721304
0.6507	0.029051	0.044293	0.07556	0.132583	0.228462	0.393894	0.800161
1.5	0.018421	0.040633	0.086698	0.171464	0.316418	0.558381	0.983627

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.010008	0.015941	0.032261	0.076031	0.191024	0.497429	1.31055
0.0231	0.010005	0.015938	0.032262	0.076032	0.191024	0.497428	1.31055
0.0531	0.013858	0.017966	0.032362	0.07603	0.191024	0.497428	1.31055
0.1225	0.022887	0.028491	0.039748	0.07703	0.191024	0.497428	1.31055
0.2823	0.03702	0.046045	0.063339	0.094633	0.193194	0.497428	1.31055
0.6507	0.060432	0.073917	0.101873	0.149214	0.231519	0.500314	1.31055
1.5	0.105871	0.123333	0.163168	0.239115	0.359006	0.578865	1.312637

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.00729	0.011203	0.021967	0.050802	0.126518	0.328254	0.863602
0.0231	0.007399	0.011204	0.021963	0.050801	0.126518	0.328254	0.863602
0.0531	0.010661	0.01381	0.022705	0.0508	0.126519	0.328254	0.863602

0.1225	0.016447	0.021124	0.030972	0.053703	0.126519	0.328254	0.863602
0.2823	0.027573	0.03433	0.048075	0.073815	0.133395	0.328254	0.863602
0.6507	0.0491	0.058782	0.078538	0.115107	0.17985	0.341163	0.863602
1.5	0.092608	0.10622	0.134547	0.18703	0.279337	0.443305	0.884307

NAND2X2

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.011518	0.014066	0.020561	0.036388	0.075779	0.179276	0.453229
0.0231	0.014501	0.016802	0.023025	0.038925	0.07866	0.182403	0.456476
0.0531	0.020782	0.023762	0.030021	0.045523	0.085394	0.189484	0.463778
0.1225	0.029486	0.03414	0.044097	0.062507	0.101844	0.206069	0.480671
0.2823	0.042133	0.049143	0.064561	0.093455	0.141801	0.245223	0.51984
0.6507	0.063268	0.073043	0.095202	0.139046	0.214717	0.338161	0.611342
1.5	0.102392	0.115662	0.146184	0.208358	0.322566	0.515424	0.826238

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.009671	0.011251	0.015326	0.025588	0.051857	0.121464	0.306023
0.0231	0.011945	0.013519	0.017646	0.028055	0.054452	0.124126	0.308712
0.0531	0.015314	0.017783	0.023202	0.033987	0.060517	0.130296	0.314933
0.1225	0.01864	0.022427	0.030736	0.046539	0.074856	0.144683	0.329376
0.2823	0.020126	0.02583	0.038423	0.062584	0.104277	0.178245	0.362794
0.6507	0.015165	0.023677	0.042478	0.078798	0.142308	0.249201	0.440583
1.5	-0.007523	0.004859	0.032516	0.086423	0.18166	0.344193	0.611881

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.008044	0.010978	0.019018	0.040586	0.09725	0.248241	0.648938
0.0231	0.008113	0.010984	0.019017	0.040586	0.097251	0.248242	0.648938
0.0531	0.012551	0.014216	0.020319	0.040581	0.097251	0.248242	0.648938
0.1225	0.02016	0.02338	0.030344	0.045568	0.097293	0.248242	0.648938
0.2823	0.032603	0.037782	0.048759	0.068633	0.109474	0.248303	0.648938

0.6507	0.053744	0.061141	0.078127	0.110318	0.16248	0.273379	0.648938
1.5	0.095719	0.104998	0.127755	0.17594	0.260551	0.393512	0.695854

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.005442	0.007192	0.012057	0.025057	0.059196	0.150142	0.391478
0.0231	0.005896	0.007424	0.012053	0.025058	0.059197	0.150142	0.391478
0.0531	0.008889	0.010718	0.014597	0.02562	0.059197	0.150142	0.391478
0.1225	0.014194	0.01676	0.022379	0.033605	0.061304	0.150142	0.391478
0.2823	0.024226	0.028018	0.036244	0.051965	0.080604	0.154694	0.391478
0.6507	0.043818	0.049294	0.061221	0.084178	0.124838	0.197607	0.399074
1.5	0.083859	0.091457	0.108572	0.141815	0.201083	0.304686	0.490046

NAND2X4

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.010572	0.011861	0.015247	0.02361	0.04372	0.095229	0.230687
0.0231	0.013669	0.014799	0.017908	0.026047	0.04634	0.098186	0.233852
0.0531	0.019648	0.021284	0.025105	0.03293	0.052931	0.105015	0.241004
0.1225	0.027873	0.030433	0.036393	0.048357	0.069785	0.121475	0.257682
0.2823	0.040304	0.044059	0.053032	0.07159	0.104723	0.161287	0.296839
0.6507	0.062383	0.067511	0.079958	0.10676	0.157438	0.244183	0.389595
1.5	0.105349	0.112177	0.128972	0.165722	0.238022	0.370328	0.588848

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.008909	0.00971	0.011813	0.017064	0.030147	0.064412	0.155046
0.0231	0.011061	0.011912	0.014009	0.01934	0.032573	0.066945	0.157629
0.0531	0.013693	0.015052	0.018271	0.024935	0.038346	0.072848	0.163616
0.1225	0.01584	0.0179	0.022797	0.032923	0.051398	0.086732	0.177555
0.2823	0.015294	0.018373	0.025691	0.040941	0.06899	0.117664	0.20998
0.6507	0.006486	0.011066	0.021949	0.044605	0.086618	0.160293	0.283764
1.5	-0.023813	-0.017217	-0.001366	0.031994	0.094211	0.204417	0.390895

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.006907	0.00835	0.012292	0.022922	0.050836	0.125223	0.322631
0.0231	0.007115	0.00834	0.012299	0.022921	0.050836	0.125222	0.32263
0.0531	0.011823	0.012659	0.015099	0.023687	0.050835	0.125223	0.322631
0.1225	0.01895	0.020721	0.024829	0.032933	0.054075	0.125223	0.322631
0.2823	0.030726	0.033585	0.040273	0.053435	0.076259	0.132834	0.322631
0.6507	0.051217	0.055126	0.064854	0.085748	0.122682	0.182417	0.337092
1.5	0.093168	0.09783	0.110041	0.138606	0.19582	0.292804	0.444255

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.00487	0.005738	0.008102	0.014518	0.031321	0.076071	0.194816
0.0231	0.005546	0.006243	0.00831	0.014517	0.031321	0.076071	0.194816
0.0531	0.008282	0.009291	0.011723	0.016663	0.031572	0.076072	0.194816
0.1225	0.013143	0.014567	0.017926	0.024903	0.038413	0.077052	0.194817
0.2823	0.022407	0.024481	0.029357	0.039433	0.058147	0.093288	0.196604
0.6507	0.040979	0.043906	0.050823	0.065191	0.092189	0.140739	0.231597
1.5	0.080056	0.083883	0.093346	0.11365	0.152033	0.222066	0.344803

NOR2X1

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.02137	0.025868	0.037554	0.068163	0.148092	0.360811	0.92055
0.0231	0.024598	0.029095	0.040862	0.071569	0.151561	0.36431	0.924176
0.0531	0.031429	0.035988	0.04776	0.078542	0.158628	0.37143	0.931556
0.1225	0.041349	0.047611	0.061748	0.09293	0.173123	0.386028	0.946712
0.2823	0.053314	0.063255	0.083571	0.123029	0.205206	0.41838	0.980172
0.6507	0.063544	0.079781	0.112796	0.171443	0.274916	0.492152	1.055872
1.5	0.073217	0.097138	0.147023	0.238238	0.390042	0.65665	1.227442

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.029987	0.034515	0.046144	0.075723	0.151386	0.351233	0.880382

0.0231	0.032127	0.036708	0.048464	0.078162	0.153901	0.353782	0.882949
0.0531	0.037903	0.042552	0.054462	0.084388	0.16032	0.360309	0.889524
0.1225	0.052737	0.057676	0.06948	0.099509	0.175682	0.375864	0.905178
0.2823	0.076101	0.083454	0.100573	0.135705	0.211778	0.412058	0.941488
0.6507	0.10732	0.118229	0.14396	0.197374	0.295627	0.496396	1.025577
1.5	0.14825	0.164023	0.20167	0.281381	0.430821	0.69193	1.220877

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.014199	0.019936	0.035889	0.078782	0.191332	0.491075	1.286442
0.0231	0.014237	0.01995	0.035895	0.078782	0.191333	0.491075	1.286442
0.0531	0.015123	0.020375	0.035986	0.078785	0.191333	0.491075	1.286442
0.1225	0.021595	0.026424	0.039539	0.079398	0.191333	0.491075	1.286442
0.2823	0.035857	0.040863	0.05469	0.089748	0.193115	0.491075	1.286442
0.6507	0.065972	0.071739	0.087683	0.125876	0.217224	0.493959	1.286442
1.5	0.11984	0.126848	0.1481	0.1979	0.30084	0.545006	1.28962

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.018292	0.023733	0.03805	0.075562	0.173525	0.434142	1.125228
0.0231	0.018278	0.023735	0.038051	0.075563	0.173525	0.434142	1.125228
0.0531	0.018151	0.023685	0.038046	0.075562	0.173526	0.434142	1.125228
0.1225	0.022998	0.027417	0.039854	0.075574	0.173526	0.434142	1.125228
0.2823	0.037359	0.0438	0.057662	0.08667	0.174544	0.434142	1.125228
0.6507	0.061197	0.071207	0.092662	0.132827	0.206545	0.436653	1.125228
1.5	0.100819	0.116247	0.149524	0.212059	0.317771	0.50999	1.128062

NOR2X2

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.019392	0.021718	0.027685	0.042959	0.082421	0.187164	0.46492
0.0231	0.022668	0.024969	0.030962	0.046323	0.085876	0.190672	0.468453
0.0531	0.029531	0.031941	0.037968	0.053339	0.092978	0.197859	0.475683
0.1225	0.039103	0.042613	0.050661	0.068021	0.107753	0.21275	0.490663

0.2823	0.050401	0.056182	0.068831	0.093324	0.14031	0.245884	0.524038
0.6507	0.060498	0.069848	0.090542	0.129888	0.197391	0.320971	0.600278
1.5	0.072193	0.08587	0.116603	0.176791	0.281821	0.457588	0.774297

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.026957	0.029244	0.035146	0.049922	0.086964	0.183665	0.438718
0.0231	0.02913	0.031445	0.037429	0.052328	0.089473	0.186238	0.441322
0.0531	0.034989	0.037334	0.0434	0.058472	0.095836	0.192764	0.447931
0.1225	0.049587	0.052245	0.058679	0.073637	0.11115	0.208308	0.463632
0.2823	0.071373	0.075312	0.084994	0.106234	0.147598	0.244652	0.500089
0.6507	0.099498	0.105364	0.11992	0.152242	0.215573	0.330055	0.584801
1.5	0.133862	0.142372	0.163658	0.211635	0.307683	0.483643	0.78276

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.011811	0.014593	0.022385	0.043442	0.098784	0.246091	0.636879
0.0231	0.011854	0.014633	0.022395	0.043443	0.098784	0.246091	0.636879
0.0531	0.012968	0.015423	0.022701	0.043486	0.098784	0.246091	0.636879
0.1225	0.019465	0.021812	0.028146	0.046136	0.098888	0.246091	0.636879
0.2823	0.033786	0.036126	0.042936	0.06086	0.10672	0.246364	0.636879
0.6507	0.063646	0.06628	0.074229	0.094728	0.142141	0.263893	0.637032
1.5	0.117039	0.119973	0.130156	0.158006	0.218159	0.345224	0.671879

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.015397	0.018103	0.025186	0.043408	0.09047	0.215368	0.546392
0.0231	0.015395	0.018109	0.02518	0.043406	0.09047	0.215368	0.546392
0.0531	0.015272	0.017973	0.025144	0.043406	0.09047	0.215369	0.546392
0.1225	0.020699	0.023064	0.028777	0.04479	0.090471	0.215369	0.546392
0.2823	0.034139	0.037716	0.045921	0.062481	0.099021	0.21541	0.546392
0.6507	0.056583	0.062177	0.075049	0.100776	0.146426	0.239636	0.546419
1.5	0.094104	0.10278	0.12284	0.163257	0.234824	0.354261	0.599248

NOR2X4

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.018215	0.019394	0.022453	0.030145	0.049606	0.100867	0.236557
0.0231	0.021507	0.022667	0.025709	0.033428	0.052981	0.104329	0.240065
0.0531	0.02823	0.029529	0.032691	0.040422	0.05999	0.111431	0.247246
0.1225	0.037285	0.039191	0.04375	0.053761	0.074752	0.126254	0.262187
0.2823	0.04775	0.050907	0.058375	0.073737	0.102536	0.159307	0.295541
0.6507	0.05781	0.062813	0.074776	0.099684	0.144845	0.223349	0.371608
1.5	0.072769	0.079956	0.097298	0.134069	0.203012	0.323032	0.524876

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.024292	0.025403	0.028305	0.035572	0.053432	0.099182	0.218761
0.0231	0.026434	0.027561	0.030513	0.037872	0.05585	0.101688	0.221317
0.0531	0.032241	0.033373	0.036355	0.043804	0.061961	0.107996	0.227753
0.1225	0.045975	0.047344	0.050802	0.058823	0.076896	0.123092	0.243043
0.2823	0.065069	0.067082	0.072251	0.084327	0.109548	0.158801	0.278695
0.6507	0.088342	0.091307	0.098969	0.117095	0.15546	0.229592	0.362194
1.5	0.113933	0.118201	0.129281	0.155718	0.212603	0.325027	0.525734

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.010509	0.011845	0.015601	0.02581	0.052779	0.12459	0.314987
0.0231	0.010555	0.011894	0.015625	0.02582	0.052778	0.12459	0.314987
0.0531	0.01189	0.013014	0.016339	0.02604	0.052797	0.12459	0.314987
0.1225	0.0184	0.019492	0.022678	0.030868	0.054694	0.124601	0.314987
0.2823	0.032772	0.033842	0.037041	0.045952	0.0685	0.129924	0.314988
0.6507	0.062205	0.063362	0.067027	0.077533	0.103035	0.162679	0.326089
1.5	0.11404	0.115196	0.119502	0.133437	0.16822	0.241532	0.399205

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.013406	0.014708	0.018113	0.026828	0.048917	0.107111	0.261166
0.0231	0.013406	0.014691	0.018124	0.026829	0.048917	0.107111	0.261166

0.0531	0.013357	0.014636	0.018021	0.026789	0.048916	0.10711	0.261167
0.1225	0.019328	0.020567	0.023595	0.030591	0.050099	0.107112	0.261166
0.2823	0.031939	0.033828	0.038439	0.048424	0.067606	0.113798	0.261167
0.6507	0.052895	0.055836	0.063052	0.078701	0.108427	0.160797	0.278988
1.5	0.088369	0.092902	0.104051	0.128346	0.174869	0.256528	0.391065

MAJ3X1

Cell Rise Delay (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.023603	0.027902	0.038986	0.065929	0.133903	0.312956	0.787095
0.0231	0.024703	0.029254	0.04038	0.06793	0.136311	0.315565	0.789761
0.0531	0.030598	0.034467	0.045233	0.072657	0.141697	0.32152	0.795994
0.1225	0.041591	0.046948	0.058862	0.085508	0.154733	0.335196	0.810118
0.2823	0.055303	0.063089	0.081031	0.117134	0.186976	0.36726	0.842621
0.6507	0.072374	0.083447	0.109631	0.163451	0.260894	0.444306	0.918611
1.5	0.094593	0.109897	0.146239	0.222832	0.36751	0.620451	1.09853

Cell Fall Delay (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.024064	0.026914	0.036121	0.058937	0.11688	0.270449	0.677493
0.0231	0.025228	0.029111	0.038317	0.061655	0.120082	0.273843	0.680993
0.0531	0.030737	0.034247	0.043768	0.067518	0.126717	0.281003	0.688412
0.1225	0.04095	0.046061	0.057037	0.080953	0.14071	0.295753	0.703617
0.2823	0.053495	0.060477	0.07702	0.110478	0.173387	0.328924	0.737425
0.6507	0.067137	0.077521	0.101851	0.151764	0.242364	0.406735	0.81532
1.5	0.08007	0.095141	0.130237	0.202748	0.337349	0.572692	0.997344

Rise Transition Time (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.026575	0.031258	0.044465	0.080373	0.174062	0.423457	1.084952
0.0231	0.025951	0.030933	0.044303	0.080125	0.173864	0.423282	1.084816
0.0531	0.025998	0.030419	0.043785	0.080044	0.173854	0.423239	1.084771
0.1225	0.03564	0.038658	0.048487	0.08046	0.17384	0.423243	1.084751

0.2823	0.04821	0.054321	0.068971	0.097869	0.177097	0.423233	1.08475
0.6507	0.074162	0.082762	0.103157	0.144494	0.220353	0.430742	1.084745
1.5	0.124615	0.136129	0.16422	0.22376	0.332583	0.526969	1.096581

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.01946	0.023868	0.034303	0.063261	0.139039	0.340823	0.876292
0.0231	0.019147	0.023671	0.034453	0.063181	0.138927	0.340828	0.876297
0.0531	0.019728	0.023546	0.034259	0.063162	0.138956	0.340832	0.876283
0.1225	0.027302	0.030361	0.038446	0.063536	0.138811	0.340798	0.87627
0.2823	0.037859	0.043482	0.055636	0.080405	0.142798	0.34067	0.876252
0.6507	0.060657	0.067675	0.085187	0.121309	0.185677	0.350549	0.876162
1.5	0.106836	0.116987	0.140701	0.190231	0.2836	0.448121	0.894182

MAJ3X2

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.021209	0.023025	0.029151	0.042817	0.07653	0.163097	0.391299
0.0231	0.022828	0.024618	0.030492	0.044396	0.078626	0.165609	0.393961
0.0531	0.028778	0.030597	0.035801	0.04927	0.083593	0.171285	0.400149
0.1225	0.039489	0.042181	0.048899	0.063241	0.096758	0.184823	0.414328
0.2823	0.053399	0.057392	0.067369	0.089085	0.130631	0.2178	0.447363
0.6507	0.072251	0.077965	0.092236	0.124081	0.186349	0.297943	0.526012
1.5	0.10012	0.10801	0.127762	0.172192	0.26183	0.428896	0.71461

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.021245	0.022802	0.026377	0.037153	0.064048	0.13392	0.318701
0.0231	0.022673	0.024112	0.028478	0.039636	0.066729	0.137029	0.322011
0.0531	0.027744	0.029302	0.033584	0.045099	0.072757	0.143694	0.329054
0.1225	0.036682	0.038879	0.045026	0.058211	0.086096	0.157667	0.343646
0.2823	0.046024	0.049578	0.058563	0.078288	0.11636	0.189969	0.376406
0.6507	0.054722	0.060092	0.073162	0.102293	0.159138	0.261467	0.452866
1.5	0.057858	0.065565	0.084627	0.126727	0.209645	0.362381	0.624499

Rise Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.023882	0.026041	0.032487	0.049485	0.094509	0.214256	0.531467
0.0231	0.022759	0.025594	0.032216	0.049367	0.09445	0.214092	0.531331
0.0531	0.02338	0.025729	0.031681	0.048991	0.094396	0.21409	0.531294
0.1225	0.033244	0.035078	0.039278	0.05268	0.094299	0.21405	0.531287
0.2823	0.044111	0.047574	0.055204	0.073547	0.107932	0.214892	0.531268
0.6507	0.069242	0.0735	0.084615	0.109323	0.156787	0.248584	0.532866
1.5	0.116981	0.122834	0.137814	0.172227	0.241341	0.365558	0.60529

Fall Transition Time (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.017126	0.018746	0.023604	0.03648	0.070594	0.161531	0.402897
0.0231	0.016875	0.018246	0.023508	0.03656	0.070368	0.161459	0.402903
0.0531	0.017482	0.018802	0.023605	0.036376	0.070348	0.161392	0.402908
0.1225	0.02463	0.026103	0.030637	0.040532	0.07062	0.161311	0.402883
0.2823	0.034704	0.037688	0.044236	0.058742	0.086102	0.163697	0.402796
0.6507	0.05572	0.059771	0.069346	0.089971	0.130116	0.202656	0.408299
1.5	0.100226	0.105395	0.118319	0.146678	0.203556	0.308221	0.494767

MAJ3X4

Cell Rise Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.020762	0.02204	0.024764	0.031894	0.049482	0.093841	0.208358
0.0231	0.022128	0.023052	0.025515	0.033211	0.05139	0.096078	0.210881
0.0531	0.028235	0.029066	0.031584	0.038478	0.056169	0.101165	0.216688
0.1225	0.038626	0.039982	0.043658	0.052074	0.069702	0.114297	0.2303
0.2823	0.052278	0.05439	0.059701	0.072369	0.098502	0.148058	0.263031
0.6507	0.071842	0.074713	0.082257	0.100275	0.138653	0.212675	0.343328
1.5	0.102817	0.106776	0.117086	0.141756	0.19505	0.30215	0.496979

Cell Fall Delay (ns)							
tr \ Cload	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758

0.01	0.020153	0.021151	0.023323	0.028174	0.04205	0.076652	0.167706
0.0231	0.021813	0.022612	0.02456	0.030476	0.044302	0.079459	0.170637
0.0531	0.026808	0.027557	0.029642	0.035487	0.049664	0.085462	0.177219
0.1225	0.034526	0.035856	0.038868	0.046685	0.062546	0.098522	0.19094
0.2823	0.042039	0.043993	0.048647	0.06001	0.083833	0.129234	0.222282
0.6507	0.047646	0.050362	0.057246	0.073681	0.108535	0.175869	0.295496
1.5	0.044784	0.048616	0.058658	0.082394	0.132417	0.230295	0.407663

Rise Transition Time (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.022866	0.023706	0.027287	0.035994	0.058315	0.118099	0.27646
0.0231	0.021553	0.023169	0.026695	0.035105	0.057889	0.117877	0.276352
0.0531	0.02233	0.023691	0.02666	0.034401	0.057656	0.117864	0.276324
0.1225	0.032112	0.033606	0.035718	0.041317	0.059977	0.117733	0.276292
0.2823	0.042773	0.044831	0.048548	0.059342	0.081626	0.127033	0.276241
0.6507	0.067102	0.069128	0.075148	0.08951	0.119626	0.177174	0.299032
1.5	0.113748	0.116675	0.124497	0.143669	0.186008	0.269393	0.416085

Fall Transition Time (ns)							
tr \ Cloud	0.0005	0.0013	0.0035	0.0094	0.0249	0.0662	0.1758
0.01	0.016329	0.017346	0.019607	0.026036	0.042632	0.087397	0.206082
0.0231	0.016305	0.017126	0.018889	0.025586	0.042697	0.087284	0.206117
0.0531	0.016972	0.017572	0.019509	0.025533	0.042551	0.087273	0.206072
0.1225	0.023679	0.024862	0.026924	0.032575	0.045561	0.087051	0.206028
0.2823	0.033649	0.03508	0.0387	0.046811	0.064743	0.099493	0.20654
0.6507	0.053857	0.055677	0.060933	0.072616	0.097763	0.145977	0.237398
1.5	0.09681	0.099247	0.106168	0.122111	0.156147	0.224675	0.348246